

Measurement of the Higgs boson production in the diphoton decay channel

by

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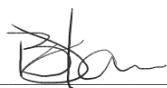
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I, the undersigned, Bongiwe Dlamini a student of BACHELOR OF SCIENCE hereby declare the project work presented in this report is my own work and has been carried out under the supervision of Dr Shell-May Liao of the University of Eswatini.

This work has not been previously submitted to any other university for any examination.



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Abstract

Measurements of Higgs boson mass is carried out using pp collisions data at $\sqrt{s} = 13$ TeV with an integrated luminosity of $5.3fb^{-1}$ recorded by the ATLAS experiment at the CERN Large Hadron Collider. This study reports evidence of Higgs boson production via the diphoton decay channel. The analysis is based on variation of cuts on transverse momentum, isolation energy and pseudorapidity of photon candidates. These cuts are used to improve how well we can spot real Higgs events by reducing background noise and making the signal clearer. Results obtained from the combinations of these variations are presented. However, there were minor deviations observed in some datasets which are maybe due to data quality and modelling inaccuracies. Despite slight deviations observed in some datasets, most of the measurements are consistent with expectations for the Standard Model Higgs boson mass.

Acknowledgements

I would like to thank my supervisor Dr Shell-May Liao for the support she gave me to make this work a success, her insightful answers to even my basic questions and for her valuable input every time. I would also like to thank my colleague Nhlanhla for the brainstorming whenever help was needed and for the casual conversations that enhanced my understanding of the concept.

Dedication

I dedicate this thesis to my mother, Jabulile. For everything you have done and continue to do for me mother, I will forever be grateful.

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Chapter 1

Introduction

On July 4th, 2012, the search for the Higgs boson culminated with the announcement that ATLAS (A Toroidal LHC Apparatus) and CMS(Compact Muon Solenoid) had independently discovered a new particle with a mass close to 125 GeV [1]. The discovery was unambiguous, with signals detected in multiple channels, demonstrating high statistical significance [2]. This new particle had 0 spin, no electric charge, even parity, width, cross-sections, branching ratios, signal strength, and couplings [3]. This was subsequently confirmed to match the expected properties of the Higgs boson.

1.1 Basics and Theory

1.1.1 Background

Previously, CMS observed an excess of events above the expected background, with a local significance of 5.0 standard deviations [1], at a mass near 125 GeV, signalling the production of a new particle. The expected significance for a standard model (SM) Higgs boson of that mass is 5.8 standard deviations. The excess was most significant in the two decay modes with the best mass resolution, gg and ZZ ; a fit to these signals gave a mass of 125.3 ± 0.4 (stat.) ± 0.5 (syst.) GeV. The decay to two photons indicated that the new particle is a boson with a spin different from one.

After conducting yet another experiment [4], CMS performed an extensive analysis on the full dataset collected in 2011 and 2012. Reconstruction of this data was done using final detector calibration values pp collisions at the LHC. Integrated luminosity was 5.1fb^{-1} at a centre-of-mass energy of 7 TeV (herein referred to as the “7 TeV dataset”) and 19.7fb^{-1} at 8 TeV. The mass of the Higgs boson was measured to be $124.70 \pm 0.34\text{ GeV}$.

In 2015 ATLAS Collaboration and CMS Collaboration published a new measurement of m_H using a combination of the Run 1 data from ATLAS and CMS experiments, leading to improved precision for m_H [5]. And recently CMS published a paper where m_H measurement with 2016 data set, in the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ \rightarrow 4\ell$ decay channels, was combined with measurement of m_H with the Run 1 data set. When the Run 2 results with the 2016 data set was combined with the same measurement performed in Run 1 at 7 and 8 TeV, finding that $m_H = 125.38 \pm 0.14\text{ GeV}$. This is claimed to be currently the most precise measurement of the mass of the Higgs boson.

1.1.2 Theoretical foundations

The ATLAS detector

The Large Hadron Collider (LHC) is the the largest and highest-energy particle accelerator. It is installed in Geneva-Switzerland in an underground tunnel which was initially the Large Electron Positron collider (LEP) [6]. The LHC is a 27.6 km circumference of two beam pipes that cross at four interaction regions. The LHC comprises of eight straight sections each approximately 500 metres in length available for experimental insertions. It was designed to allow particles of the same charge: protons or heavy ions – to be accelerated and collided with a centre of mass energy of 14 TeV and an unprecedented luminosity of $10^{34}\text{cm}^{-2}\text{s}^{-1}$. The fundamental design parameters of the accelerator are the maximum energy and luminosity reachable. High rate of event production is about equally important, to allow for a high flux of particles resulting in a sufficiently high number of collisions.

The ATLAS electromagnetic (EM) calorimeter is primarily composed of liquid argon as the ionizing medium and lead as the absorber material. Its design utilizes an accordion geometry, which enhances the granularity of the detector and improves energy resolution. This configuration allows for precise localization of energy deposits, which is essential for

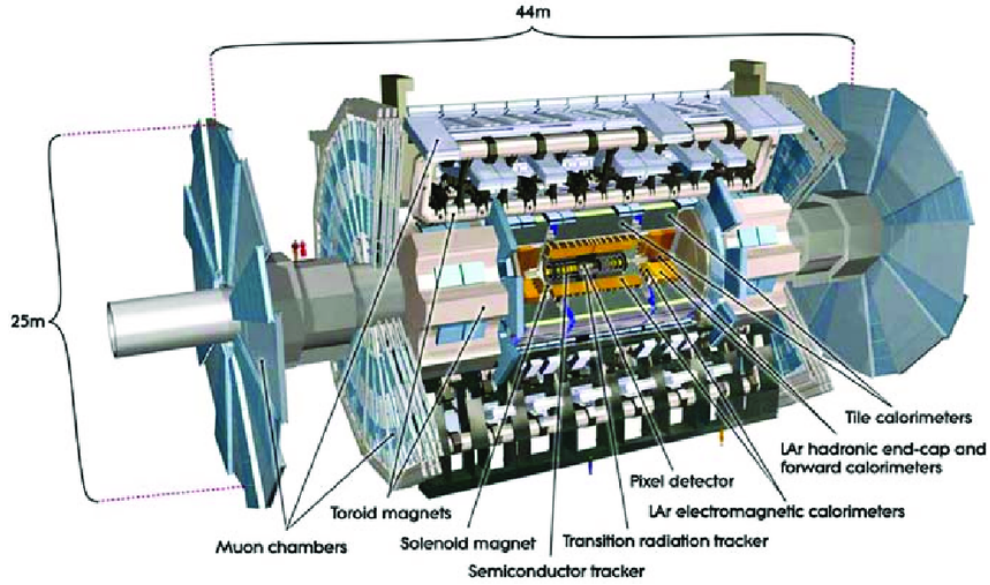


Figure 1.1: The ATLAS detector at the Large Hadron Collider, a general-purpose particle detector designed to observe and analyze proton-proton collisions to explore fundamental particles and forces.

accurately reconstructing the trajectories of charged particles and identifying photons.

The calorimeter operates by absorbing the energy of incoming particles [7]. When a charged particle or photon interacts with the lead, it produces ionization in the liquid argon. The resulting ionization electrons are collected by highly granular readout electrodes, which convert the ionization into measurable electrical signals. This process enables the calorimeter to provide both energy measurements and spatial information about the incident particles. Figure 1.1 shows the ATLAS detector.

Higgs boson production

There are four main ways to produce the Higgs boson; through the fusion of two gluon particles (gluon-fusion, or ggF), through the fusion of weak vector bosons (VBF), in association with a W or Z boson (VH), and in association with one or more top quarks (ttH+tH). All these are best explained using the Feynmann diagrams also called loop diagrams.

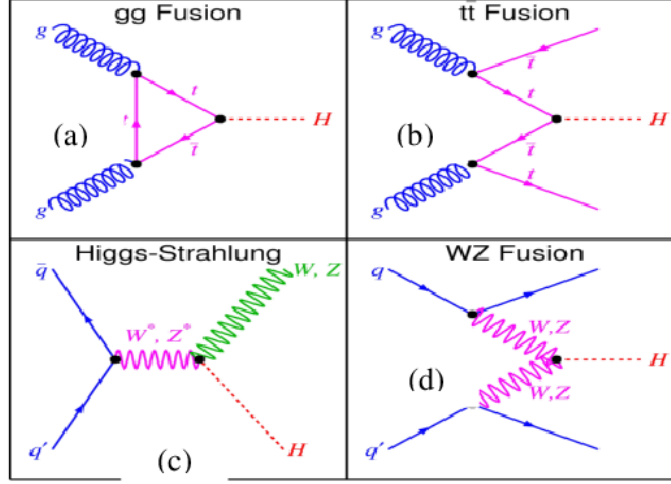


Figure 1.2: Feynman diagram illustrating Higgs boson production

Since the Higgs boson is unstable it can decay in different ways [8]. Among these can be decays into gluons, fermions, photons, gauge bosons and sometimes supersymmetric decay. Of interest in this paper is the decay of the Higgs boson into photons which is mediated by W bosons and heavy fermion loops in the SM [9]. The Higgs boson decays into photons [10] through a process involving loop diagrams, where the Higgs boson first decays into a virtual charged particle-antiparticle pair, and these particles then annihilate, producing photons.

Loop diagrams in Higgs decays describe special processes where the Higgs boson decays into particles like photons, even though it doesn't interact with them directly. Instead, the decay happens through a "loop" of virtual particles-like W bosons or heavy quarks-that briefly pop in and out of existence. These virtual particles form a loop in the Feynman diagram, helping the Higgs boson turn into two photons. This type of decay, called a loop-induced decay, is important because it allows scientists to study the Higgs boson using photons, which are easier to detect. In contrast, decays like $H \rightarrow ZZ \rightarrow 4$ (four leptons) happen at tree level, meaning the Higgs directly decays into Z bosons, which then decay into leptons without needing loops. This decay is also clean and provides detailed information about the Higgs, but it's less common because it requires enough energy to produce two massive Z bosons.

The Higgs mechanism is a fundamental process in the Standard Model of particle physics. It explains how fermions and gauge particles acquire their mass [11]. It is closely associated with the Higgs field and the Higgs boson. The Higgs field is a scalar field that permeates all of space. It has a nonzero value everywhere, even in otherwise empty space. This nonzero value breaks the weak isospin symmetry of the electroweak interaction. A crucial phenomenon of the Higgs mechanism is symmetry breaking [12] and mass generation. In the SM, the Higgs field interacts with gauge bosons (specifically the W and Z bosons) and fermions. The mass of a particle is determined by the strength of its interaction with the Higgs field [13].

Simply put, electroweak symmetry breaking is a process that helps explain why some particles, like electrons and W/Z bosons, have mass. It happens because of the Higgs field. Normally, we'd expect this field to be zero, but it actually has a non-zero value everywhere—called the vacuum expectation value (VEV). This constant background value “breaks” the original symmetry of the electroweak force, causing particles that interact with the Higgs field to gain mass. Without this process, everything would remain massless, and atoms wouldn't form the way they do.

The reason why the Higgs mechanism works lies in the so-called “Mexican Hat potential” of the Higgs field. The ground state of the Higgs field is not at the centre of the potential but reaches its minimum at non-zero field values. As nature has to choose one of the lowest energy states, the potential is not symmetric with respect to that state (this is called spontaneous symmetry breaking) [14]. The Mexican hat has the minimum necessary functions: a metastable point at the center of the hat for high energies, and a lower stable position. The potential energy when the field has any given magnitude has a graph which looks like a Mexican hat, which gives the model its name. The Mexican hat potential is caused by the Higgs field interacting with itself. This interaction creates a non-zero vacuum expectation value, which gives particles their mass.

1.1.3 Problem Statement

The contribution of the Higgs boson to the mechanism of electroweak symmetry breaking [15] is yet a mystery. Clarifying the role of the Higgs boson in electroweak symmetry breaking lies in studying its properties, in particular its mass. However, due to the ever escalating amounts of data acquired as a result of increase in luminosity and energy levels of the proton-proton collisions, the focus has shifted from mere observation to precision

measurements of the Higgs properties. Besides its intrinsic importance as a fundamental parameter, improved knowledge of m_H yields more precise predictions for the other Higgs boson properties [16]. This motivates precise measurements of the mass of the Higgs boson m_H in all available decay channels.

1.1.4 Objectives

This paper provides a detailed report on pp collisions by jet reconstruction in the diphoton decay channel. The aim is to rediscover the Higgs boson with an improved mass measurement. Diphoton invariant mass will be measured to determine if any Higgs bosons are produced. To achieve this the following will be done;

- collecting data sample
- writing a code that will simulate proton-proton collisions
- presenting-graphically-the results of the collisions
- providing a comprehensive interpretation of the results

1.1.5 Justification

Compared to previous experiments conducted on the same study using Run 1 data from the LHC, this study explores Run 2 data from LHC data with relatively higher luminosity and high center of mass energy $\sqrt{s} = 13TeV$. Additionally, this experiment benefits from the technological advancements and upgrades made to the LHC during the Long Shutdown 1 (LS1) from 2013 to early 2015. These repairs and changes provide an advantageous data set compared to Run 1 data. [17]. Even though there are current Higgs mass measurements obtained from Run 3 data but, LHC produces about one billion collisions per second meaning tonnes of data are produced. It is therefore of paramount importance that every single bit of this data is fully explored to optimize the effectiveness of LHC as a whole.

This paper is outlined in the following way. Chapter 2 section 1.1 gives details about the data sample used in this study, section 1.2 dives into event selection, and event reconstruction is discussed in section 1.3. Similarly, Chapter 4 has sections; 3.1 on signal modelling, 3.2 on background modelling while section 3.3 describes in detail the variation

of cuts. Results and their discussion are provided in Chapter 4, while Chapter 5 concludes the study.

Chapter 2

Event selection and reconstruction

This chapter provides an overview of the data sample used in this experiment, and focuses on the process of event selection and event reconstruction. It begins by describing the data sample and continues to delve into the event selection and reconstruction techniques employed to accurately identify and measure particles resulting from the pp collisions.

2.1 Data

This study explores proton-proton collisions data at $\sqrt{s} = 13$ TeV recorded by Run 2 of the LHC(2015-2018). But this paper uses a 2015 data sample. The data samples are those recorded by the LHC upon satisfying data quality requirements [18]. There are three data sets used in this paper, data A, data and data C.

The uncertainty in the combined 2015-2018 integrated luminosity is 1.7% [19] Integrated luminosity of the 2015 data is 3.24 fb^{-1} with a total uncertainty of 0.07 % [20]. Due to the low peak luminosity ($0.5 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ in 2015) [21], the number of proton-proton interactions occurring in the same bunch crossing is low, on average $\langle\mu\rangle = 13.4$.

However, Run 2 of the LHC was not limited to the data sample used in this experiment. In 2016 the intergrated lumunosity of the data was 32.9 fb^{-1} while the pile up was $\langle\mu\rangle = 25.1$. In 2017 43.9 fb^{-1} , and the corresponding pile up was $\langle\mu\rangle = 37.8$. Lastly the 2018 data sample had an intergrated lumunosity of 58.5 fb^{-1} with a pile-up of $\langle\mu\rangle = 36.1$ [21]

Pile-up happens when many proton collisions occur at the same time in the detector. This adds extra, unwanted energy around particles like photons, making it harder to tell if a photon is truly isolated or just surrounded by noise from other collisions. When isolation is affected, background might be wrongly identify as real Higgs signals. It also makes it harder to measure the photons' energy and direction accurately, which distorts the precision of the diphoton invariant mass-the key value that is equivalent to the Higgs.

Primary diphoton and single-photon triggers are used to efficiently select events with Higgs boson candidates in the diphoton decay channel. During 2015 both the primary single-photon and diphoton trigger utilized loose identification requirements [21]. The single-photon trigger's E_T thresholds was increased from 120 to 140 GeV in 2016. Events of Higgs boson candidates in diphoton decay channels are mainly selected with the diphoton trigger whose E_T thresholds are 35 and 25 GeV for the leading and subleading photon respectively allowing the collection of diphoton events with invariant masses low enough for good background modelling for resonances above 120 GeV.

2.2 Event selection

As mentioned above, event selection necessitates the presence of at least two photons satisfying the loose identification criteria. Transverse momentum (p_T) and transverse energy (E_T) being the main quantities of the photons considered, it is required that the two highest- p_T preselected photons are designated as the candidates for the diphoton system. These photon candidates are required to be in a fiducial region of the EM calorimeter defined by $|\eta| < 2.37$, [22] excluding the transition region between the barrel and the end cap calorimeters ($1.37 < |\eta| < 1.52$). This region takes two photon candidates arranging them according to their E_T where the leading and subleading photon candidates are required to have $E_T/m_{\gamma\gamma} > 0.35$ and 0.25 , respectively, where $m_{\gamma\gamma}$ is the invariant mass of the two selected photons. Below is an equation for the the invariant mass of the two photons:

$$m_{\gamma\gamma} = \sqrt{2E_1E_2(1 - \cos\alpha)}. \quad (2.1)$$

In this equation E_1 and E_2 are the energies of the leading and the subleading photon, while α is the opening angle between these two photons with respect to the their production vertex.

A neural network algorithm identifies the diphoton primary vertex [23]. This is fundamental for the resolution of the α measurement and therefore enhances the $m_{\gamma\gamma}$ measurement. This neural network algorithm takes in information about the reconstructed vertices in the event and the trajectories of the two photons measured using the depth segmentation of the calorimeter and completed by photon conversion information if present. Event selection required the total mass of the two photons to lie within 105 and 160 GeV.

2.3 Event reconstruction

Dynamical, topological cell-clustering algorithm is used to form energy deposits in the calorimeter which are then used to reconstruct photons. Dynamic clusters change in size as needed to recover energy from electrons from photon conversions [24]. A photon candidate is classified as either converted or unconverted. It is converted if it is matched to either two tracks forming a conversion vertex, or one track with the signature of an electron track without hits in the innermost pixel layer- otherwise, it is classified as unconverted.

Photon conversion classification involves identifying when a photon changes into an electron-positron pair, usually after interacting with material in a detector. This process is important because it helps in accurately tracking and identifying particles. By classifying photon conversions, we can better separate real photons from background noise, improving the accuracy of measurements like energy and direction. Techniques often use the paths (tracks) of the electron and positron to locate the conversion point. Understanding and classifying these conversions helps improve data quality.

Photon candidates are differentiated from jet backgrounds through identification criteria that rely on variables related to the calorimeter shower shape. A loose selection criteria is applied during the preselection stage, and the final photon candidate selection is made using stricter criteria called the tight selection. The efficiency of the tight identification for reconstructed photon candidates varies from approximately 84% (85%) at $p_T = 25$ GeV to 94% (98%) for unconverted (converted) photons with $p_T > 100$ GeV [24].

The final photon selection incorporates both calorimeter and track-based isolation criteria to effectively reduce the contamination from jets that may be misidentified as photons. The calorimeter isolation is calculated by summing the total energy of the calorimeter clusters in a cone of size $\Delta R = 0.2$ around the photon candidate. Similarly, the track-based isolation is defined as the scalar sum of the transverse momenta of tracks within the same cone size. Only tracks with p_T greater than 1 GeV are considered for this isolation, and

these tracks must be associated with the primary vertex of the selected diphoton event, excluding those linked to photon conversion vertices if present. Furthermore, each photon candidate must meet a track isolation criterion, requiring that the track isolation be less than 5% of the photon's transverse energy to ensure a cleaner photon sample.

Chapter 3

Signal modelling, background modelling and variation of cuts

In this chapter we first provide a brief overview of signal and background modelling. We then explore in detail the impacts of varying cuts on transverse momentum, energy isolation and pseudorapidity in the transition region.

3.1 Signal modelling

This study considers events with diphoton final states for a Higgs boson with mass $m_H = 125\text{GeV}$. The invariant mass of any parent particle susceptible to decay into two daughter particles, is related to properties of the daughter particles through the formula:

$$M^2 = E^2 - p^2, \tag{3.1}$$

where E is the total energy of the daughter particles, and p is the the magnitude of the vector sum of the momenta of the daughter particles.

In this analysis the Gaussian function is used to model the shape of the resonance (Higgs boson peak) in the diphoton invariant mass distribution. The Gaussian function commonly used [25] is :

$$f(m_{\gamma\gamma}) = A \exp\left(-\frac{(m_{\gamma\gamma} - m_0)^2}{2\sigma^2}\right) \quad (3.2)$$

where A is the normalization factor (amplitude of the peak), $m_{\gamma\gamma}$ is the diphoton invariant mass, m_0 is the mean central value of the peak (125 GeV), σ is the standard deviation (width of the Gaussian).

3.2 Background modelling

This is an essential procedure in the analysis which enables isolation of Higgs signal from background noise like photon-photon scattering and jet to photon misidentification. It encompasses estimating the contribution from processes that are not related to the Higgs boson but can mimic its signature in the detectors. In this analysis a 4th-order polynomial is used to model the background signal. The polynomial function models background contributions using bin centers as input values. This is a basic 4th-order polynomial:

$$f(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4, \quad (3.3)$$

capable of describing more complex shapes in the distribution of background events that might not be captured by the Gaussian model. The coefficients of the polynomial can be calculated for each event category (Higgs production mode/decay channel) using a simultaneous fit to the sets of simulated signal mass resonances [26]. The Higgs signal is extracted once the background is modeled, by subtracting the background model from the data, hence revealing the characteristic resonance associated with the Higgs boson. In some cases other analytic functions used to model the background signal include power-law functions and Bernstein polynomials [27].

Systematic uncertainties in background estimation come from factors that can affect how accurately we predict background events, such as detector performance, simulation models, and how well we understand particle interactions. These uncertainties can change

the shape or size of the background, making it harder to separate signal from noise. To reduce their impact, control regions are used, data-driven methods, and cross-checks with simulations. Carefully studying these uncertainties helps ensure that any signal seen is real and not just due to a mistake or bias in the background prediction.

3.3 Cuts variation

This section outlines the exploitation of cuts. In particle physics a cuts refer to a specific selection criteria applied to the data to isolate events of interest and reduce background noise. In particular these criteria define the kinematic and detector acceptance regions, this is to ensure that only events meeting certain conditions are analyzed. There are a number of cuts in the context of pp collisions. These may include cluster quality cuts and shower shape cuts [28]. But we discuss only the cuts on transverse momentum and isolation energy and pseudorapidity because these are the main contributors to the $m_{\gamma\gamma}$ as shown by equation 3.1.

3.3.1 Transverse momentum (p_T)

As mentioned, this experiment explores the cut on transverse momentum of the photons. Transverse momentum is a crucial observable for understanding the fundamental quantum chromodynamic (QCD) interactions involved in proton-proton collisions [29]. It is calculated as the projection of particle's momentum onto a plane perpendicular to the beam direction. Applying a cut on p_T ensures that only events that come from the Higgs decay are selected, therefore improving signal-to-background ratio. It also helps to reduce background i.e low- p_T photons, can often come from non-Higgs sources, like jets misidentified as photons. Hence imposing a minimum p_T cut we reduce these misidentifications.

In Higgs boson measurements, transverse momentum (pT) thresholds are used to select events where particles have enough sideways energy to be interesting. Setting these thresholds helps reduce background noise from low-energy particles that are less likely to come from a Higgs decay. However, if the threshold is too high, some real Higgs events with lower pT particles might be missed, reducing the data sample. If it's too low, more background events get included, making it harder to find the Higgs signal. Choosing the

right p_T threshold balances keeping true Higgs events and rejecting background for better measurement accuracy.

The initial p_T values for the leading (sub-leading) photon candidates are 40(30) GeV. The variation that will be employed in this cut is random, but based on the theory which make predictions of the diphoton invariant mass on certain thresholds. This exploration will be carried out on the individual data samples A,B, and C as long as on the combination of these, which therefore make a fourth data sample.

In high-energy physics, transverse momentum refers to the component of a particle's momentum perpendicular to the beam axis. Since the incoming protons in LHC experiments move along the longitudinal axis (the z-axis), transverse momentum provides a frame-independent observable that reflects the dynamics of the underlying partonic interaction. High p_T signatures typically arise from hard scatter processes-such as quark-gluon interactions, Higgs boson production, or the decay of heavy resonances. Conversely, low p_T particles are commonly associated with soft QCD interactions and underlying event noise, which are generally of lesser interest in searches for new physics. Because of this, transverse momentum is an ideal variable for discriminating between interesting signal events and background noise. Requiring particles or jets to have p_T above certain thresholds enhances the purity of the selected event sample and optimizes computational resources [30].

The ATLAS experiment employs a sophisticated trigger system to filter events in real-time. Given the LHC's collision rate of 40 MHz, it is infeasible to store all collision data. The trigger system reduces the data volume by a factor of approximately 10 000, recording only about 1 kHz of the most promising events [31].

During Run 2, the trigger system comprised of two levels: Level-1 (L1), a hardware-based system using calorimeter and muon chamber data, and the High-Level Trigger (HLT), a software-based system capable of near-full event reconstruction. Transverse momentum cuts were central to both levels of this system. At L1, object identification such as electrons, photons, and jets was performed with reduced granularity, applying p_T thresholds of 20–100 GeV depending on the object and trigger path. The HLT refined these selections using full detector information, with variable p_T thresholds optimized for each physics channel [32].

p_T cuts are motivated by the need to increase the signal-to-background ratio. Standard Model processes like QCD multi-jet production dominate the LHC cross section, producing

a profusion of low-energy particles. These backgrounds can shade the occurrence of rare processes such as electroweak boson production or decays of hypothetical particles like supersymmetric states or dark matter candidates.

For example, in supersymmetry searches, events are often required to have jets with $p_T > 100$ or 200 GeV and significant missing transverse energy (E_T^{miss}) due to the presence of weakly interacting particles [33]. Applying these cuts significantly reduces backgrounds from QCD multijet events, which rarely produce high p_T and significant E_T^{miss} simultaneously.

3.3.2 Isolation Energy

In high-energy physics, energy isolation discusses the requirement that a photon (or other particle) be separated from other nearby particles in terms of energy deposition. The isolation energy is generally defined as the sum of the transverse energies (E_T) deposited by other particles within a cone of a certain radius $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ around the candidate particle direction, excluding the particle itself [34].

In Higgs boson measurements, isolation energy helps identify particles from the Higgs decay by measuring nearby energy. However, pile-up-many overlapping collisions in the same event-adds extra energy around particles, making isolation less accurate. Isolation energy corrections remove this extra pile-up energy, so we can better separate real Higgs decay particles from unrelated activity. Correcting for pile-up improves the accuracy of particle identification and reduces background noise, helping to make clearer and more precise measurements of the Higgs boson properties.

For photon identification in the ATLAS experiment, we say a photon is "isolated" if the transverse energy deposited within a cone of radius $\Delta R = 0.4$, not associated with the photon shower itself, is below a specified threshold-usually a few GeV [35]. This cut helps us distinguish prompt photons from photons originating in hadronic jets, which tend to be embedded in larger deposits of calorimetric energy.

The ATLAS detector is a general-purpose detector designed to observe a wide range of particles which are produced in proton-proton collisions. It consists of several sub-detectors arranged in a cylindrical geometry around the interaction point, including the inner tracking detector, the electromagnetic and hadronic calorimeters, and the muon spectrometer as discussed in chapter1.

For the $H \rightarrow \gamma\gamma$ channel, the most relevant components are the inner detector and the electromagnetic calorimeter (ECAL). The ECAL is a high-granularity, lead-liquid argon sampling calorimeter designed to measure the energy and direction of electrons and photons with high precision. Photon candidates are reconstructed from clusters of energy deposited in the ECAL and are classified as either "converted" or "unconverted" depending on whether they are associated with tracks in the inner detector from photon conversions.

The isolation energy cut relies on the calorimeter's ability to accurately measure the transverse energy deposited by other particles in the vicinity of the photon candidate. AT-

LAS uses both calorimeter and track-based isolation criteria. The calorimetric isolation is corrected for the photon energy leakage and pile-up contributions using sophisticated algorithms developed during Run 2.

During Run 2 the LHC operated at a center-of-mass energy of 13 TeV with increasing luminosity, leading to higher levels of pile-up-multiple overlapping pp collisions in a single bunch crossing. This increased the complexity of photon isolation, as pile-up can contribute fake energy deposits in the isolation cone, mimicking non-isolated photons.

To address this, ATLAS applied pile-up subtraction techniques to the isolation variables. A dynamic estimate of the pile-up contribution to the isolation energy was calculated event-by-event and subtracted from the measured value. Isolation cuts were typically applied to both the calorimeter and track-based isolation:

- **Calorimetric Isolation:** We note that the total transverse energy in a cone of $\Delta R = 0.4$ around the photon, excluding the core of the photon cluster, had to be below a threshold-typically around 6 GeV [36].
- **Track Isolation:** It was a necessity that the scalar sum of the transverse momenta of tracks with $p_T > 1$ GeV within a smaller cone of $\Delta R = 0.2$ around the photon be below a few GeV [36].

These thresholds were optimized based on Monte Carlo simulations and data control regions to ensure high photon identification efficiency while suppressing backgrounds.

The precision measurement of the Higgs boson mass in the diphoton channel relies heavily on the purity of the selected photon sample. The primary backgrounds in this channel are:

- **Prompt diphoton production** from QCD processes.

- **Photon-jet and dijet events** where jets are misidentified as photons.

Isolation cuts are crucial for reducing the second type of background. Fake photons originating from jets are generally surrounded by hadronic activity, which increases the isolation energy. By rejecting such candidates, the analysis improves the signal-to-background ratio, which is vital for reconstructing a clean invariant mass peak for the Higgs boson [36].

Furthermore, accurate isolation helps reduce systematic uncertainties related to photon identification and energy scale calibration. Since the Higgs mass is extracted from the invariant mass distribution of the two photons, any bias or inefficiency in photon selection can directly affect the measurement. In the ATLAS Run 2 measurement, the Higgs boson mass in the diphoton channel was determined with an uncertainty of about 0.3 GeV, a precision that would not be possible without very strict isolation requirements [36].

The performance of the isolation criteria was validated using control samples such as $Z \rightarrow e^+e^-$ and $Z \rightarrow \mu^+\mu^-\gamma$ events. These samples provided clean photon-like or photon-rich environments where the isolation efficiency and energy scale corrections could be studied in detail. These studies showed that the isolation criteria retained a high efficiency (above 90%) for prompt photons while significantly reducing the rate of background events [35].

3.3.3 Pseudorapidity

In collider physics, pseudorapidity is a spatial coordinate defined as

$$\eta = -\ln \left[\tan \left(\frac{\theta}{2} \right) \right],$$

where θ is the polar angle of a particle's trajectory with respect to the beam axis. Pseudorapidity is preferred over the polar angle because differences in η are Lorentz invariant under boosts along the beam direction. It also provides a natural segmentation of the detector geometry and particle kinematics.

In the ATLAS experiments, some pseudorapidity regions-areas measuring how particles move at angles to the detector-are excluded because the detector calibration there is less reliable. These areas may have more complex detector materials or gaps, causing less accurate measurements of particle energy and direction. If data from these regions were used,

it could introduce errors or bias in the results. By excluding these problematic regions, measurements of the Higgs boson become more precise and trustworthy, improving the overall quality and reliability of the experiment’s findings.

The ATLAS electromagnetic calorimeter is divided into two main regions: the barrel $|\eta| < 1.475$ and the endcaps $1.375 < |\eta| < 3.2$ [37]. There is a transition region between the barrel and endcaps around $1.37 < |\eta| < 1.52$ where the detector response is less optimal due to structural limitations and overlapping modules. Photon identification and energy resolution are both degraded in this region, making it a critical factor in event selection.

During Run 2 of the LHC (2015–2018), the ATLAS experiment operated at a center-of-mass energy of 13 TeV, collecting data with high luminosity and pile-up conditions. For the diphoton analyses, including the Higgs mass measurement, photons were reconstructed from clusters in the electromagnetic calorimeter and categorized based on their location in η .

To ensure optimal performance, ATLAS imposed the following pseudorapidity cuts on the selected photons:

- Photon candidates must satisfy $|\eta| < 2.37$.
- Photon candidates within the calorimeter transition region ($1.37 < |\eta| < 1.52$) were excluded [38].

These criteria ensure that only photons detected in regions of the calorimeter with well-understood and stable performance are selected, thus minimizing systematic uncertainties.

The ATLAS electromagnetic calorimeter plays a central role in photon detection. It is a lead-liquid argon sampling calorimeter with fine granularity in both transverse and longitudinal directions, allowing precise energy and position measurements. The barrel section covers the range $|\eta| < 1.475$, while the endcap sections extend to $|\eta| < 3.2$. The transition region suffers from degraded energy resolution due to increased material budget and mechanical complexity [39].

Photon reconstruction begins by identifying electromagnetic energy clusters in the calorimeter. These clusters are then matched with inner detector tracks to classify photons as either converted or unconverted. Calorimeter-based variables such as shower shape, leakage into the hadronic calorimeter, and isolation energy are used to distinguish real photons from hadronic backgrounds. Because the electromagnetic calorimeter has varying performance in different η regions, the pseudorapidity-dependent calibration and identification efficiencies must be carefully accounted for [40].

The diphoton invariant mass ($m_{\gamma\gamma}$) is calculated from the energies and directions of the two selected photons. The resolution of this mass measurement is directly influenced by the energy and angular resolution of the calorimeter. Inaccuracies in photon energy measurement can smear the mass peak, reducing the precision with which the Higgs mass is extracted.

Photons detected in the transition region between the barrel and endcap calorimeters have significantly worse energy resolution due to overlapping modules, varying material thickness, and calibration uncertainties. Including such photons would degrade the $m_{\gamma\gamma}$ resolution and increase the systematic uncertainty on the mass measurement. By excluding the $1.37 < |\eta| < 1.52$ range, ATLAS ensures that only well-reconstructed photons contribute to the Higgs boson signal [38].

Moreover, the detector acceptance and efficiency are not uniform across η , which affects the signal and background modeling. The exclusion of problematic regions simplifies the modeling and reduces the risk of mismatches between simulation and data, which are crucial when extracting the Higgs mass from a fit to the diphoton invariant mass spectrum.

The pseudorapidity cut is implemented in conjunction with other selection criteria, such as transverse energy cuts ($E_T^\gamma > 35$ GeV for the leading photon and > 25 GeV for the subleading photon), photon identification, and isolation energy requirements [38]. Events are further categorized based on whether both photons fall in the barrel region (“EBEB”) or one falls in the endcap (“EBEE”). These categories have different mass resolutions and signal-to-background ratios, and are treated separately in the statistical analysis.

Events with both photons in the barrel typically offer the best mass resolution due to the superior performance of the barrel calorimeter. Consequently, these events are given higher weight in the combined fit for the Higgs mass. Excluding poorly performing η re-

Selection Cut	Variation 1	Variation 2	Variation 3	Variation 4
Photon- p_T (GeV)	(42, 30)	(40, 30)	(56, 45)	(48, 32)
Photon- η	(2, 2)	(4, 4)	(2.4, 2.5)	(3.6, 3.6)
Photon- η -cone	(1.40, 1.56)	(1.44, 1.57)	(1.37, 1.52)	(1.37, 1.52)

Table 3.1: Showing values used to vary cuts on p_T , isolation energy, and on pseudorapidity transition region.

gions enhances this category’s purity and statistical power.

In Run 2, the ATLAS collaboration used approximately 139 fb^{-1} of data to perform a precision measurement of the Higgs boson mass in the diphoton channel. The pseudorapidity cuts, along with other selection criteria, contributed to a mass measurement of

$$m_H = 125.09 \pm 0.24 \text{ GeV},$$

where the uncertainty includes both statistical and systematic components [38]. This level of precision would not have been achievable without careful control of detector performance across η , underscoring the importance of the pseudorapidity selection.

The isolation energy cut is applied to distinguish the real photons from those produced by jets. Real photon tend to have less surrounding activity(in terms of other tracks or energy deposits), while jets or fake photons have more activity surrounding them. Therefore this cut ensures that the photons used in the analysis are not contaminated.

Additionally there are certain background process e.g those produced by jet fragmentation or other hadronic activities which can potentially result in photon-like objects that are not truly photons. The energy isolation cut reduces the likelihood of these background processes contaminating photons invariant mass spectrum, thus improving the measurements of $m_{\gamma\gamma}$.

Table 3.1 shows the variations that will be applied on selection cuts. These have been selected randomly from a range of values describing the accepted ranges for the selection cuts. The variation were created as informed by the theory in section3.

Chapter 4

Results and discussion

In this chapter, we present the results from our experiment. Results were obtained using Table 3.1 However not all the variation are discussed. Instead the best two variation from Table 3.1 are taken for the analysis and compared with each other.

To make sense of our findings, we divided the results into two main sections these correspond to variation2 and 4 of Table 3.1. Each section compares the results that were obtained using these variations in of selection cuts. One section focuses on the diphoton invariant mass, which gives us a direct way to measure the Higgs boson mass. Another uses ROC (Receiver Operating Characteristic) curves, to show how well selection cuts separate real Higgs events (signal) from background events (noise).

4.1 The Diphoton invariant mass ($m_{\gamma\gamma}$)

To begin with, we present the diphoton invariant mass distributions obtained using Variation 2 in Table3.1 .Figure 4.1 shows the direct effects of these selections cuts in the mass of the Higgs boson. In the figure, (a), (b) and (c), correspond to data sets A,B, and C respectively as discussed in subsection 2.1. While (d) is a combination of all 3 data sets evaluated as a single data set. These data sets have different luminosities which make it ambiguous to compare their results, hence the focus is not on the accuracy of the m_{γ} but on the effects of the selection cuts on each data set.

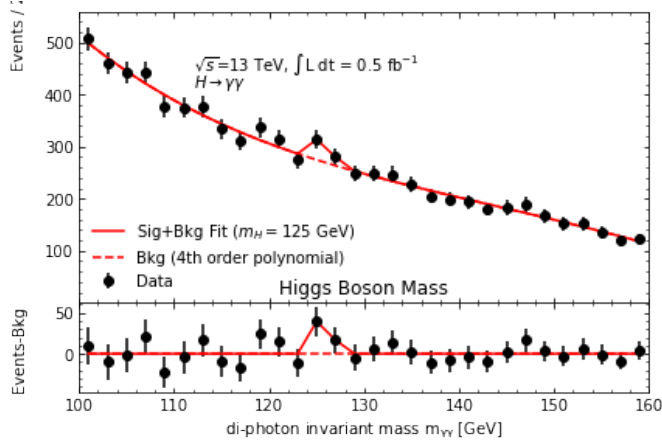
Figure 4.2 follows the same arrangement implemented in Figure 4.1, but instead uses selection cuts outlined as Variation 4 Table 3.1. We investigate the impacts of selection cuts on results shown in Figure 4.1 vs corresponding graph in Figure 4.2.

In Figure 4.1 (a), with transverse momentum cuts at 40 GeV (leading photon) and 30 GeV (subleading photon), isolation energy at 4 GeV, and pseudorapidity range $1.44 < |\eta| < 1.57$, the invariant mass is 125.93 GeV. In Figure 4.2(a), stricter cuts-transverse momentum at 48 GeV and 32 GeV, isolation energy at 3.6 GeV, and pseudorapidity range $1.37 < |\eta| < 1.52$ —result in a mass of 125.86 GeV. This was the mass closest to 125 GeV with an absolute error of 0.38% as compared to the currently accepted Higgs mass (125.38 GeV). Tighter cuts improve the mass resolution peak by reducing background noise, like jets which may be misidentified as photons. The stricter pseudorapidity cut excludes the problematic transition region, enhancing energy resolution. Higher transverse momentum and lower isolation energy cuts increase the signal-to-background ratio, making the Higgs boson peak sharper and more precise this investigation is in line with known theory [30,36].

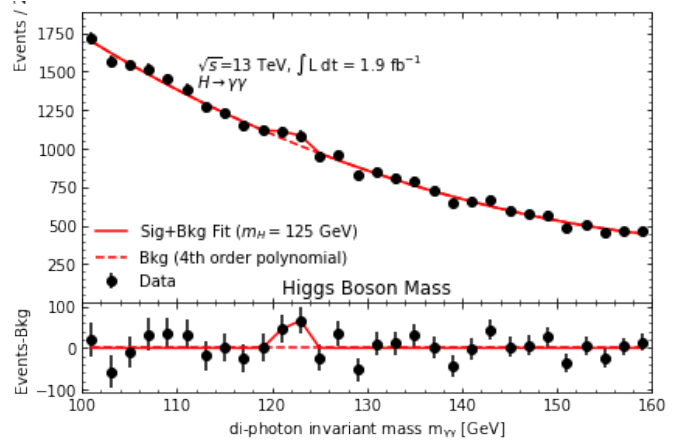
Increased luminosity leads to additional background events, making Higgs extraction more challenging, this explains why the diphoton invariant mass in Figure 4.1(b) (122.04 GeV) and Figure 4.2(b) (129.83 GeV) deviates from the expected 125 GeV, unlike Figure 4.2(a) (125.86 GeV), which is closer due to better signal clarity. In Figure 4.1(b), the cuts are looser: leading photon transverse momentum at 40 GeV, subleading at 30 GeV, isolation energy at 4 GeV, and pseudorapidity range $1.44 < |\eta| < 1.57$. Figure 4.2(b) uses stricter cuts: 48 GeV and 32 GeV for transverse momentum, 3.6 GeV for isolation energy, and $1.37 < |\eta| < 1.52$ for pseudorapidity. Even though the Higgs boson mass deviates away from the 125 GeV but these cuts clearly show how effective they can be in improving the value of the Higgs mass in simulations. The stricter cuts in Figure 4.2(b) yield a preferable mass compared to 4.1(b) because they suppress background noise and improve the signal-to-background ratio.

We expected a better diphoton invariant mass in Figure 4.2(c), but like dataset B, the mass (128.36 GeV) is slightly higher than the mass in Figure 4.1(c). As already discussed high luminosity may increase background noise making the Higgs signal hard to isolate in some cases. The selection cuts; transverse momentum, isolation energy, and pseudorapidity are the same in 4.1(c) as 4.1(a) and in 4.2(c) as 4.2(a).

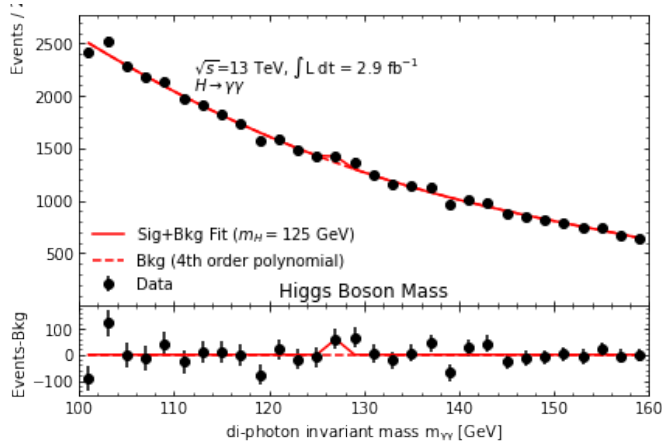
Datasets A, B, and C combined, boosts the number of Higgs events but also increases background noise as already discussed in results for Figures 4.2(b) and (c). In Figure 4.1(d),



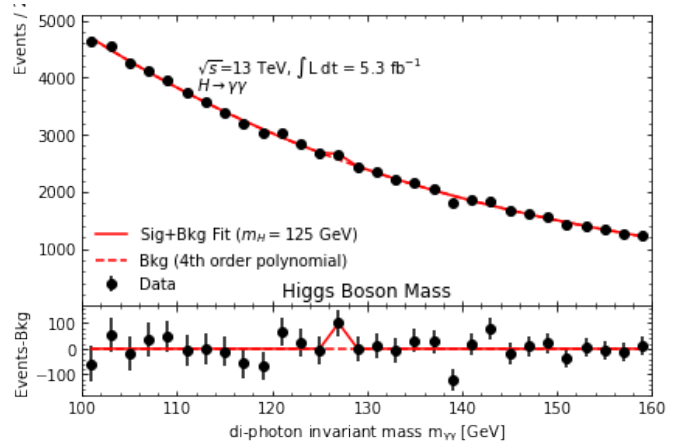
(a)



(b)

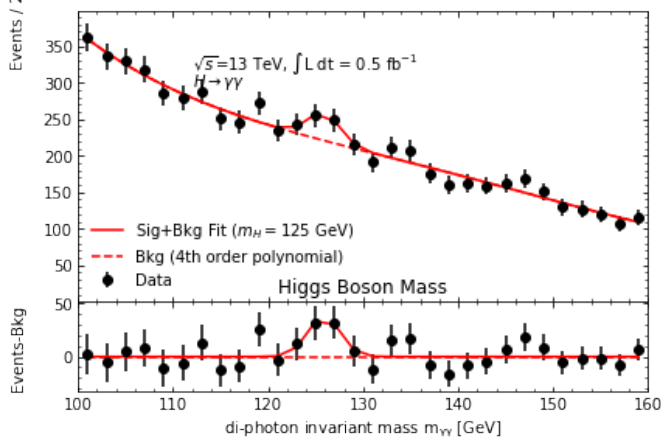


(c)

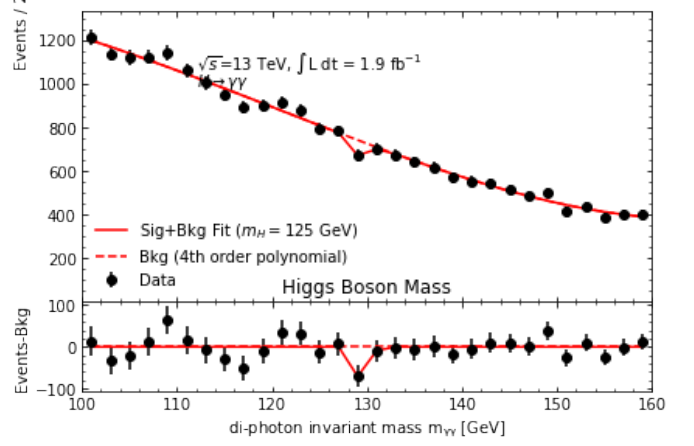


(d)

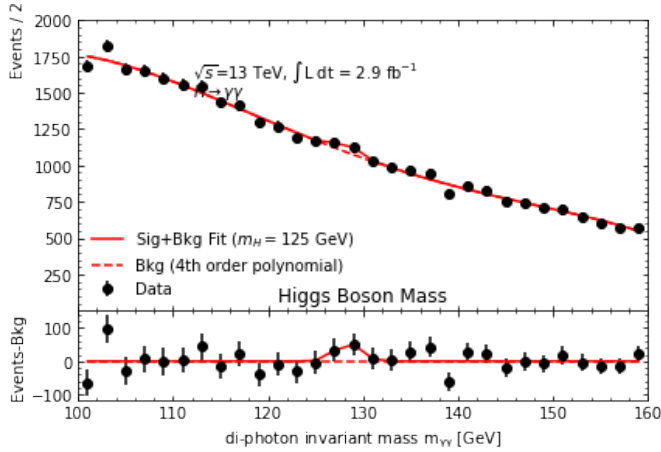
Figure 4.1: Shows diphoton invariant mass distributions simulated using Variation 2. Figures (a), (b), (c) and (d) correspond to diphoton invariant masses 125.93 GeV, 122.04 GeV, 126.58 GeV, and 127.29 GeV respectively. In this analyses events are classified as Higgs signal if they lie within 2σ of the fitted Gaussian peak centered around 125 GeV and background if they lie outside of this range or fail the cuts on photon ID, transverse momentum, isolation energy, and pseudorapidity transition regions.



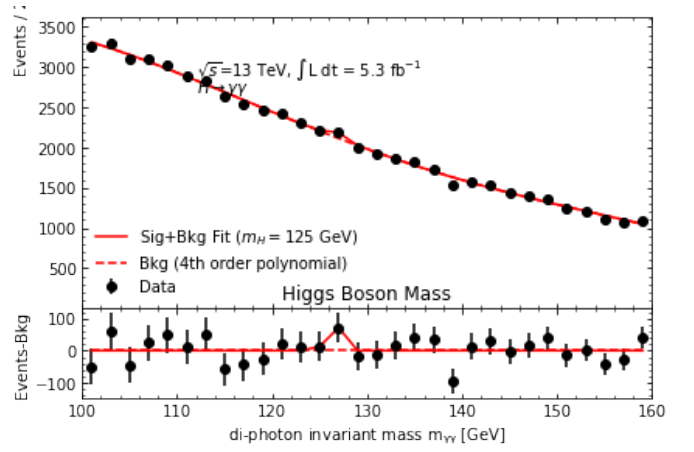
(a)



(b)



(c)



(d)

Figure 4.2: Diphoton invariant mass distributions simulated using Variation 4. Figures (a), (b), (c) and (d) correspond to diphoton invariant masses 125.86 GeV, 129.83 GeV, 128.36 GeV and 126.13 GeV and . Events lying within 2σ of the fitted Gaussian peak centered around 125 GeV are classified as Higgs signal, while those outside of this range make up background events

using less restrictive cuts (transverse momentum, isolation energy, and pseudorapidity identical to 4.1(a)), the diphoton invariant mass is 127.29 GeV. In contrast, Figure 4.2(d), with more stringent cuts (matching 4.2(d)), yields a mass of 126.13 GeV, closer to the expected 125 GeV. The precise selection criteria demonstrate a clear influence on sharpening the mass resolution peak.

The observed mass shifts are possibly a results or modelling inaccuracies, affecting the reconstructed Higgs mass peak. Pile-up effects-multiple proton-proton collisions per bunch crossing-further complicate signal reconstruction by adding noise to energy measurements and degrading resolution. Advanced techniques like pile-up mitigation algorithms, particle flow reconstruction, and vertex association are could be employed to minimize their impact. Proper treatment of both effects is crucial for accurate Higgs mass measurements and for maintaining sensitivity in precision studies and new physics searches. However this study could not include these, hence the deviations observed.

4.2 ROC curves

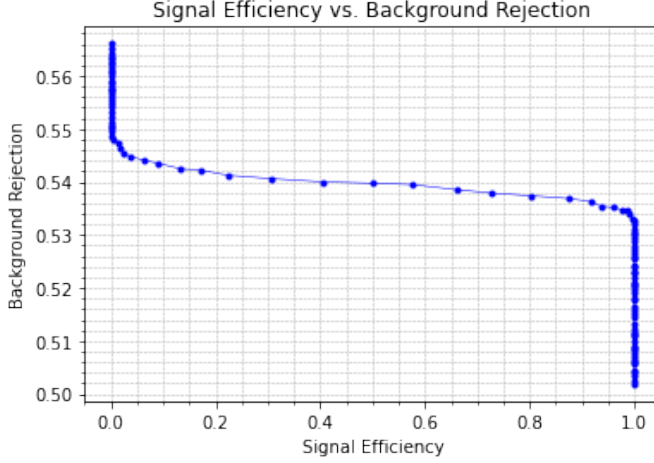
A ROC curve plots signal efficiency (true positive rate) against background rejection (1 - false positive rate) to evaluate a classifier's performance. In this experiment the classifiers are transverse momentum, isolation energy and pseudorapidity. It illustrates how well selection criteria distinguish Higgs boson signals from background noise, with higher area under the curve (AUC) indicating better

4.2.1 ROC curves 1

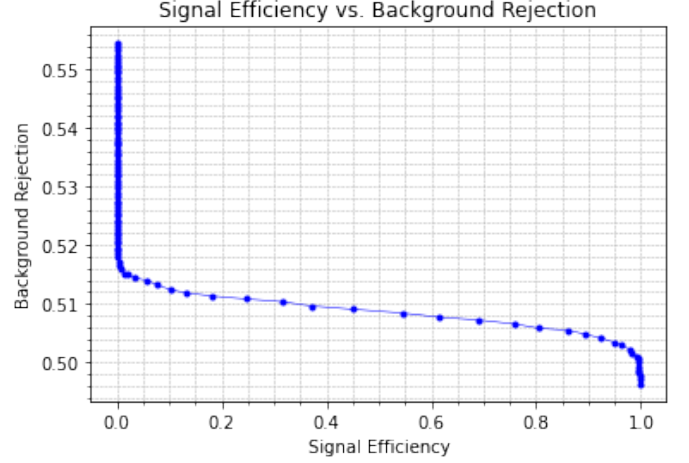
Specific values for signal rejection and background rejection extracted from Figure 4.3 are shown in Table 4.1, while table 4.2 shows those from Figure 4.4

4.2.2 ROC curves 2

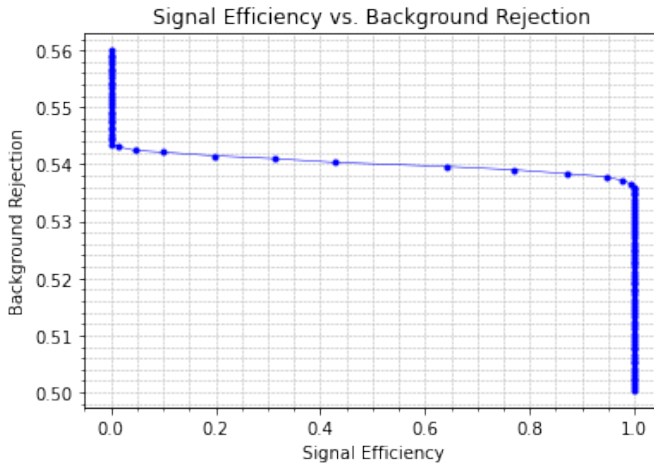
We find optimal point on the ROC curve, identifying the threshold that best balances signal efficiency (True Positive Rate, TPR) and background rejection (1 - False Positive Rate, FPR). The optimal point maximizes the trade-off between detecting Higgs boson



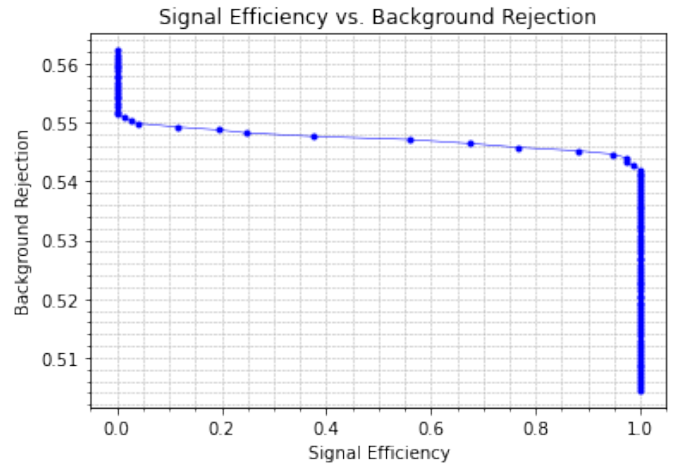
(a)



(b)

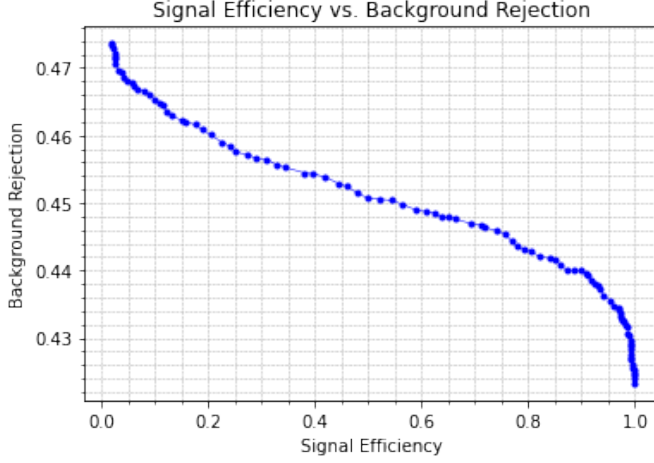


(c)

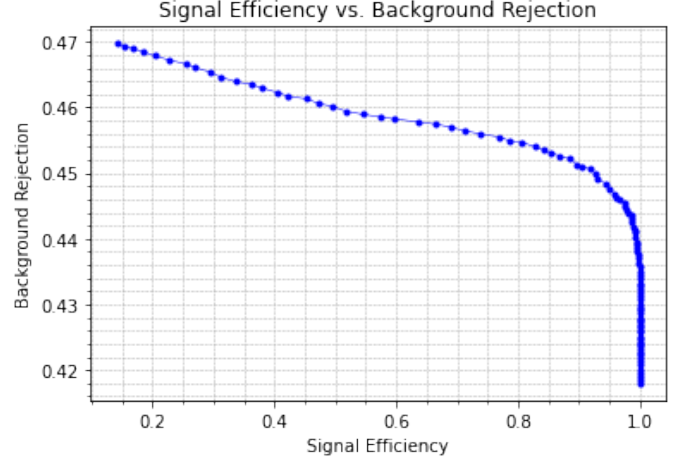


(d)

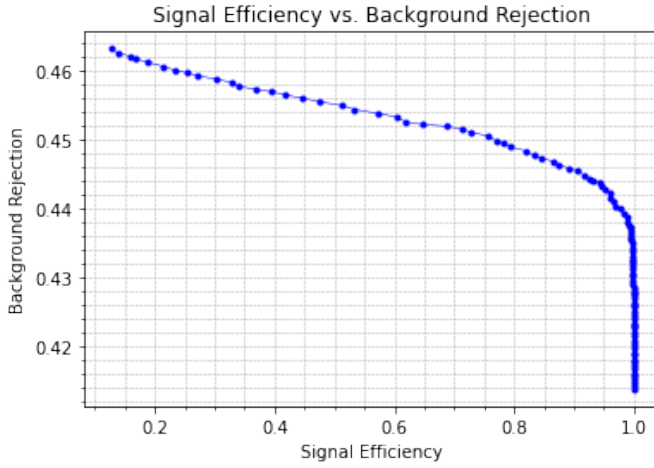
Figure 4.3: ROC curves illustrating signal efficiency versus background rejection for diphoton invariant mass distributions, simulated using varied transverse momentum, isolation energy, and pseudorapidity cuts. Graphs (a), (b), (c), and (d) correspond to diphoton invariant masses shown in Figure 4.1 (a), (b), (c) and (d) respectively, highlighting the impact of selection cuts on Higgs signal identification



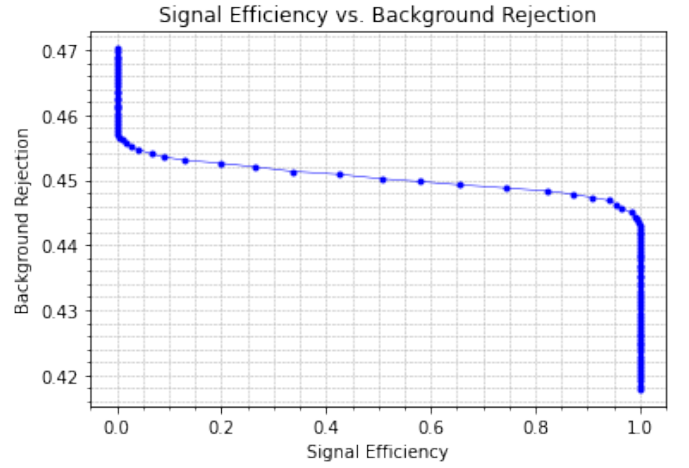
(a)



(b)



(c)



(d)

Figure 4.4: ROC curves illustrating signal efficiency versus background rejection for diphoton invariant mass distributions, simulated using varied transverse momentum, isolation energy, and pseudorapidity cuts.(a), (b), (c), and (d) correspond to Figure 4.2(a), (b), (c) and (d) respectively

Signal Efficiency	Background Rejection			
	Data A	Data B	Data C	Data A,B,C
0.2	0.542	0.515	0.542	0.549
0.4	0.540	0.510	0.541	0.548
0.6	0.539	0.508	0.540	0.547
0.8	0.538	0.506	0.539	0.546
1.0	0.52-0.534	0.460-0.500	0.500-0.536	0.504-0.542

Table 4.1: Signal efficiency versus background rejection for datasets A, B, C, and their combination, corresponding to Figure 4.1 (a), (b), (c), and (d) respectively.

Signal Efficiency	Background Rejection			
	Data A	Data B	Data C	Data A,B,C
0.2	0.460	0.468	0.461	0.452
0.4	0.454	0.462	0.457	0.451
0.6	0.449	0.458	0.435	0.450
0.8	0.443	0.454	0.449	0.448
1.0	0.423-0.430	0.418-0.440	0.412-0.438	0.418-0.444

Table 4.2: Signal efficiency versus background rejection for datasets A, B, C, and A, B, C combination corresponding respectively to Figure 4.2 (a), (b), (c), and (d).

events (signal) and minimizing background noise. One way of doing this is by the use of **Youden's J Statistic** which defines the optimal point as

$$J = TPR - FPR$$

The point with the highest J maximizes the difference between true positives and false positives. For example we can calculate J for dataA as follows:

$$J = 0.2 - 0.458 = -0.258$$

$$J = 0.4 - 0.460 = -0.060$$

$$J = 0.6 - 0.461 = 0.139$$

$$J = 0.8 - 0.462 = 0.338$$

$$J = 1.0 - 0.473 = 0.527$$

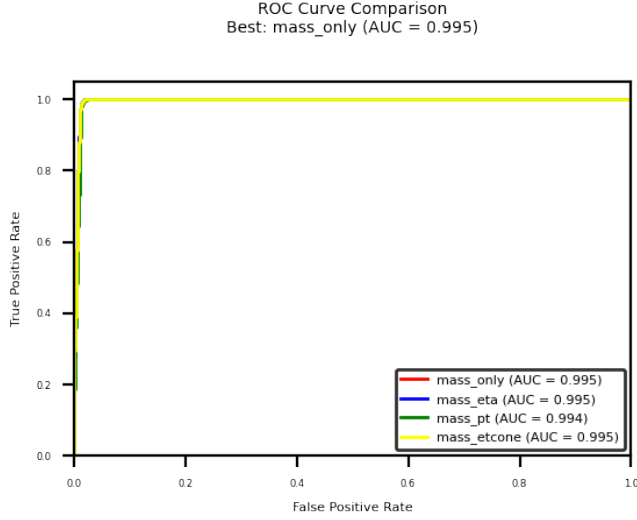
The highest $J = 0.527$ occurs at $\text{TPR} = 1.0$, $\text{FPR} = 0.473$ (background rejection = 0.527), suggesting this is the optimal point based on Youdens statistic. However, this point may not be ideal, as $\text{TPR} = 1.0$ includes all signal events but also more background (higher FPR)

For Table 4.1, linked to Figure 4.3, Youden’s J statistic identifies $\text{TPR} = 1.0$ as the optimal point for datasets A, B, C, and A,B,C, with the highest J values (e.g., 0.527 for A, 0.480 for B, 0.518 for C, 0.523 for A,B,C). However, this point includes all Higgs events but allows more background noise, leading to less precise mass measurements, like 122.04 GeV for dataset B in Figure 4.1(b). For accurate Higgs mass near 125 GeV, $\text{TPR} = 0.8$ is preferred, balancing 80% signal efficiency with strong background rejection (e.g., 0.538 for A, 0.506 for B, 0.539 for C, 0.546 for A, B, C). This aligns with sharper mass peaks, such as 125.93 GeV for A in Figure 4.1(a), improving clarity in Higgs detection.

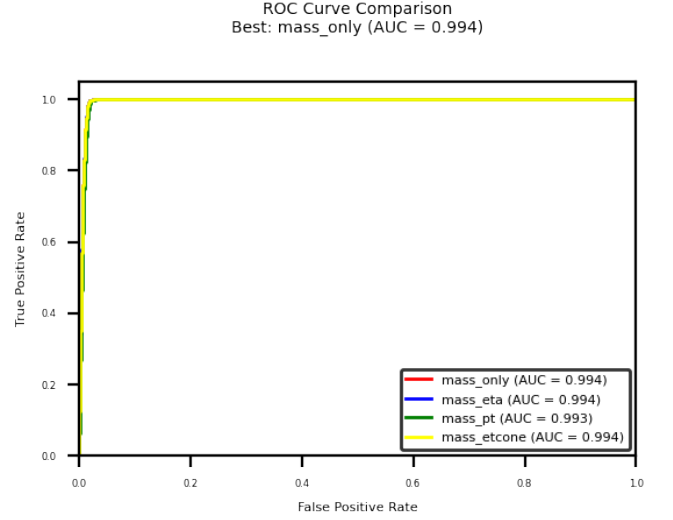
In Table 4.2, for Figure 4.4, $\text{TPR} = 1.0$ yields the highest J (e.g., 0.4265 for A, 0.429 for B, 0.425 for C, 0.431 for A, B, C), but more noise distorts masses, like 129.83 GeV for B in Figure 4.2(b). $\text{TPR} = 0.8$ is better, offering good efficiency and rejection (e.g., 0.443 for A, 0.454 for B, 0.449 for C, 0.448 for A, B, C). Stricter cuts (e.g., transverse momentum at 48 GeV, isolation energy at 3.6 GeV) produce clearer masses, like 125.86 GeV for A in Figure 4.2(a), closer to 125 GeV, by effectively reducing background noise.

Figure 4.5(a)–(d) shows ROC curves comparing how different selection cuts affect Higgs boson mass detection, with AUC values measuring how well each cut separates Higgs events from noise. The curves are named mass-only, mass-eta taken from the photon candidates [photon-eta] that gives information about isolation energy, transverse momentum photon candidates are [photon-pt] hence the name mass-pt and lastly ,mass-etcone from photon candidates [photon-etcone[20]].

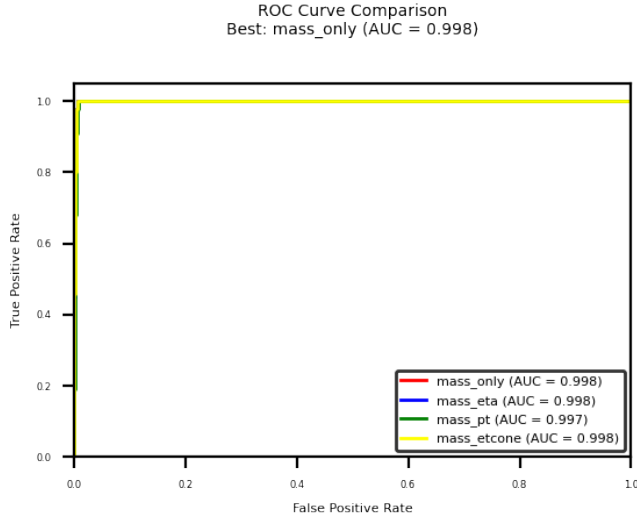
In Figure 4.5(a), mass-only, mass-eta, and mass-etcone all have $\text{AUC} = 0.995$, while mass-pt has a slightly lower $\text{AUC} = 0.994$. In Figure 4.5(b), all cuts have $\text{AUC} = 0.994$ except mass-pt at 0.993. In Figure 4.5(c) and 4.5(d), mass-only, mass-eta, and mass-etcone have $\text{AUC} = 0.998$, while mass-pt is 0.997. Higher AUC means better separation of Higgs signals. The pseudorapidity cut (mass-eta) and isolation energy cut (mass-etcone) consistently outperform the transverse momentum cut, with $\text{AUC} = 0.995$ or 0.998 across all figures. The transverse momentum cut (mass-pt) has the lowest AUC in each case. Therefore, pseudorapidity and isolation energy cuts are the most effective for accurately



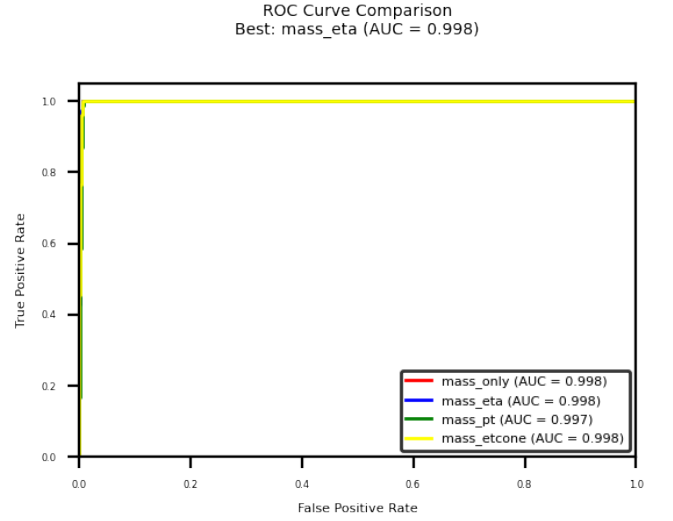
(a)



(b)

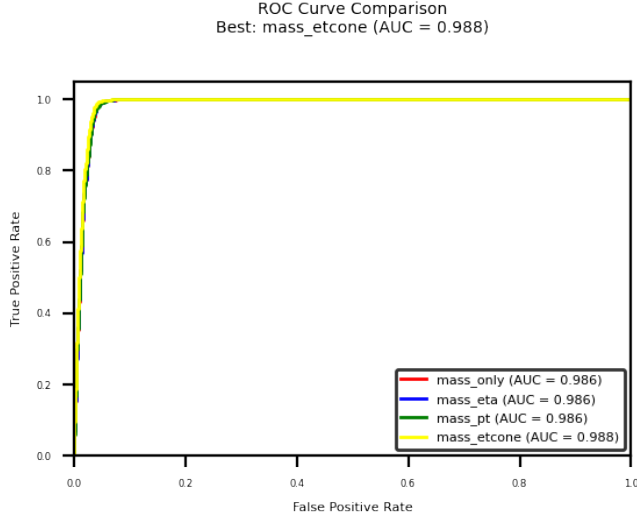


(c)

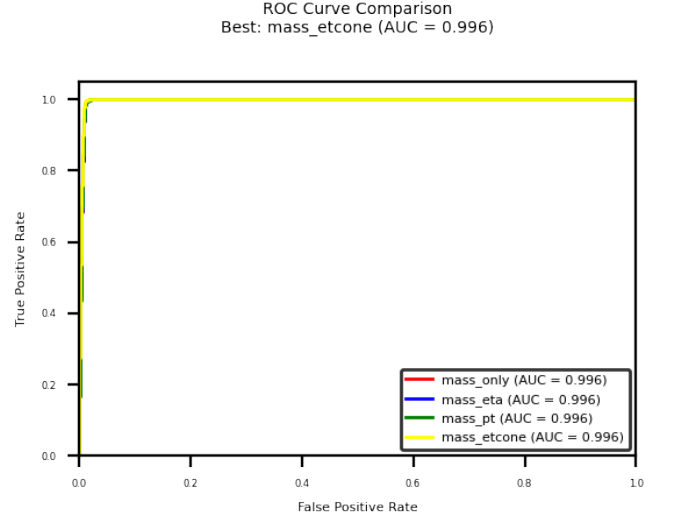


(d)

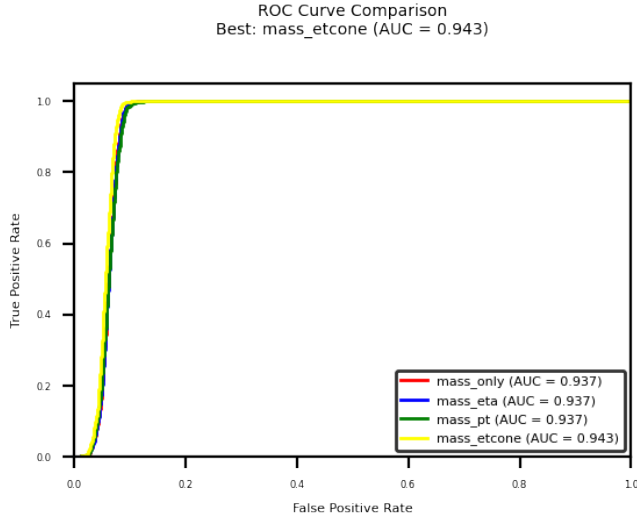
Figure 4.5: ROC curves illustrating signal efficiency versus background rejection for diphoton invariant mass distributions, simulated using varied transverse momentum, isolation energy, and pseudorapidity cuts. Graphs (a), (b), (c), and (d) correspond to datasets A, B, C, and their combination, highlighting the impact of selection cuts on Higgs signal identification



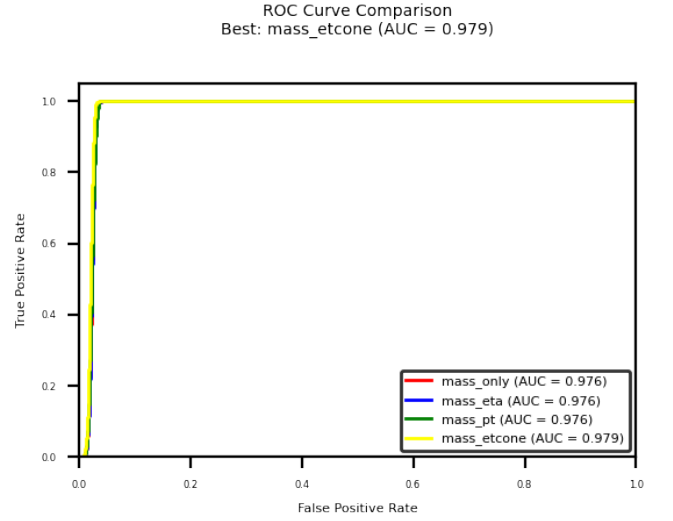
(a)



(b)



(c)



(d)

Figure 4.6: ROC curves illustrating signal efficiency versus background rejection for diphoton invariant mass distributions, simulated using varied transverse momentum, isolation energy, and pseudorapidity cuts. Graphs (a), (b), (c), and (d) correspond to datasets A, B, C, and their combination, highlighting the impact of selection cuts on Higgs signal identification

detecting the Higgs mass.

In Figure 4.6(a), mass-etcone has $AUC = 0.988$, outscoring the others at 0.986. In Figure 4.6(b), all cuts are at $AUC = 0.996$. But in 4.6(c), mass-etcone leads with $AUC = 0.943$, while the others are at 0.937. Again in Figure 4.6(d), where mass-etcone has $AUC = 0.979$, topping others at 0.976. Again we can safely conclude that the cuts on isolation energy and pseudorapidity have a significant impact in detecting the Higgs mass.

Chapter 5

Conclusion and Outlook

This study successfully confirmed the Higgs boson’s production in the diphoton decay channel. Figures 4.1 and 4.2 show diphoton invariant mass peaks around 125 GeV, for instance, 125.93 GeV for dataset A in Figure 4.1(a) and 125.86 GeV in Figure 4.2(a). These results are in line with Chapter 1 predicted Higgs mass of 125 GeV. Although there are deviations, like 122.04 GeV for dataset B in Figure 4.1(b), there are consistent peaks in datasets A, C, and A,B,C confirming Higgs detection.

It was observed that pseudorapidity and isolation energy cuts are most effective. In Figure 4.5(a)–(d), they have $AUC = 0.995$ – 0.998 , with better performance than the transverse momentum cut with $AUC = 0.993$ – 0.997 . In Figure 4.6(a), (c), (d), selection cut on pseudorapidity in the transition region performs better with $AUC = 0.988$, 0.943 , and 0.979 . These cuts are most effective in background noise reduction and mass peak sharpening and are thus best for accurate Higgs mass measurement. Future experiments may utilize these selection cuts to optimize results.

We also noticed that high luminosity aggravates the background problems. Dataset B’s mass peak (122.04 GeV in Figure 4.1(b)) shows a bigger deviation from 125 GeV, with worse background rejection (0.506 at $TPR = 0.8$ in Table 4.1). This implies high luminosity events require special attention but if applied well pseudorapidity and isolation cuts have the potential to achieve clean Higgs signals in future high-energy experiments.

The other observation is the superiority of $TPR = 0.8$ over $TPR = 1.0$. Youden’s J statistic prefers $TPR = 1.0$ (e.g., $J = 0.527$ for A in Figure 4.2(b)), but is noisier, with

mass values way greater/lower than 125 (e.g., 129.83 GeV for dataset B in Figure 4.2(b)). TPR = 0.8 provides a good compromise between 80 signal efficiency and good rejection (e.g., 0.538 for A in Figure 4.2(b)), resulting in sharper peaks like 125.93 GeV.

This analyses provides first hand information proving the existence of the Higgs boson. Even though this measurement has been carried out by previous experiments, but a lot remains to be uncovered about the Higgs boson. This demands that the experiment be carried out at even higher luminosities. This paper has provided clear insights on the potential of certain selections cuts, in particular, isolation energy and pseudorapidity cuts to improve the Higgs resonance.

In a Higgs boson experiment using the diphoton decay channel, applying optimized selection cuts on p_T , isolation energy, and pseudorapidity is crucial for enhancing mass resolution and signal significance. Raising the threshold helps suppress background from soft processes, while strict isolation cuts reduce contamination from jets misidentified as photons. Pseudorapidity cuts limit detector edge effects and improve photon energy resolution. These refined selections sharpen the invariant mass peak, allowing for more precise Higgs mass measurements.

Future high-energy experiments can further improve Higgs measurements by employing advanced machine learning techniques for event classification, and improved pile-up mitigation strategies to maintain mass precision under higher luminosities.

APPENDICES

Appendix A

A.1 Python code for making Higgs boson plots

<https://github.com/Bongiwe-Dlamini/Measurement-of-the-Higgs-boson-mass/blob/main/mass.ipynb>

A.2 Python code for ROC curves 1

https://github.com/Bongiwe-Dlamini/Measurement-of-the-Higgs-boson-mass/blob/main/roc_curve1.ipynb

A.3 Python code for ROC curves 2

https://github.com/Bongiwe-Dlamini/Measurement-of-the-Higgs-boson-mass/blob/main/roc_curve2.ipynb

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